

```
In [1]: # =====  
# Notebook setup  
# =====  
  
%load_ext autoreload  
%autoreload 2  
  
# Control figure size  
figsize=(15, 4)  
  
from util import util  
import os
```

Remaining Useful Life

Remaining Useful Life

The *Remaining Useful Life* is a key concept in predictive maintenance

The RUL refers to *the time until a component becomes unusable*

- If we can estimate the RUL of a component
- ...We can schedule maintenance operations only when they are needed

Current best practices are based on *preventive maintenance*

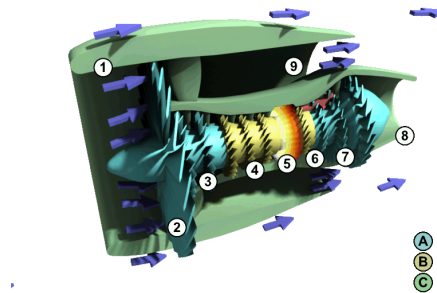
I.e. on having a *fixed maintenance schedule* for each component family

- RUL prediction can lead to significant savings
- ...By delaying maintenance operations w.r.t. the schedule
- ...But only as long as we are still able to prevent critical failures

The Dataset

We will consider the NASA [C-MAPSS dataset](#)

- The Modular Aero-Propulsion System Simulation (MAPSS)
- ...Is a NASA-developed simulator for turbofan engines



- It comes with both a Military (MAPSS) and commercial versionn (C-MAPSS)
- They different in the attributes of the considered engines

The Dataset

The C-MAPSS system can simulate a number of faults and defects

...And it was used to build a high-quality dataset for the PHM08 conference

- The dataset consists of 4 "training set" files and 4 "test set" files
- The *training set* files contain multiple run-to-failure experiments
- The *test set* files contain truncated experiments

PHM-08 hosted a competition based on this dataset

The goal was to predict the RUL at the end of each truncated experiment

- This is fine as long as the focus is on pure prediction
- ...But we want to tackle *the whole predictive maintenance problem*

As a consequence, we will focus only on the "training" data

The Dataset

Each training file refes to different faults and operating conditions

Dataset	Operating conditions	Fault modes
FD001	1 (sea level)	HPC
FD002	6	HPC
FD003	1 (sea level)	HPC, fan
FD004	6	HPC, fan

Fault modes refer to degration of either:

- The High Pressure Compressor

- The fan at the "mouth" of the engine

Inspecting the Data

Let's have a look at the row data

```
In [2]: data_folder = os.path.join('.', 'data')
data = util.load_data(data_folder)
data.head()
```

```
Out[2]:
```

	src	machine	cycle	p1	p2	p3	s1	s2	s3	
0	train_FD001	1	1	-0.0007	-0.0004	100.0	518.67	641.82	1589.70	1400
1	train_FD001	1	2	0.0019	-0.0003	100.0	518.67	642.15	1591.82	1400
2	train_FD001	1	3	-0.0043	0.0003	100.0	518.67	642.35	1587.99	1400
3	train_FD001	1	4	0.0007	0.0000	100.0	518.67	642.35	1582.79	1400
4	train_FD001	1	5	-0.0019	-0.0002	100.0	518.67	642.37	1582.85	1400

5 rows x 28 columns

- Columns "p1, p2, p3" refer to controlled parameters
- Columns "s1" to "s21" refer to sensor reading
- Binning has already been applied in the original dataset

Statistics

Let's check some statistics

```
In [3]: dt_in = list(data.columns[3:-1]) # Exclude metadata
data[dt_in].describe()
```

Out [3]:

	p1	p2	p3	s1	s2
count	160359.000000	160359.000000	160359.000000	160359.000000	160359.000000
mean	17.211973	0.410004	95.724344	485.840890	597.361022
std	16.527988	0.367938	12.359044	30.420388	42.478516
min	-0.008700	-0.000600	60.000000	445.000000	535.480000
25%	0.001300	0.000200	100.000000	449.440000	549.960000
50%	19.998100	0.620000	100.000000	489.050000	605.930000
75%	35.001500	0.840000	100.000000	518.670000	642.340000
max	42.008000	0.842000	100.000000	518.670000	645.110000

8 rows × 24 columns

There are no missing values:

```
In [4]: data[dt_in].isnull().any().any()
```

Out[4]: np.False_

Heatmaps

Let's prepare for displaying all time series

First, we standardize each column:

```
In [5]: data_sv = data.copy()
data_sv[dt_in] = (data_sv[dt_in] - data_sv[dt_in].mean()) / data_sv[dt_in].s
```

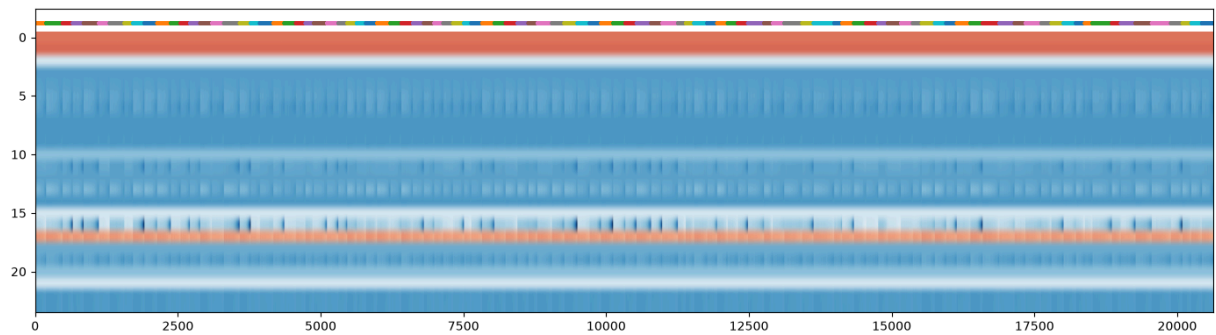
Then, we split our data based on the source file:

```
In [6]: data_sv_dict = util.split_by_field(data_sv, field='src')
print('{{{}}}'.format(', '.join(f'{k}: ...' for k in data_sv_dict.keys()))
{train_FD001: ..., train_FD002: ..., train_FD003: ..., train_FD004: ...}
```

Heatmaps

Now, let's plot all parameters and sensors for *FD001*

```
In [7]: tmp = data_sv_dict['train_FD001']
util.plot_dataframe(tmp[dt_in], labels=tmp['machine'], figsize=figsize)
```

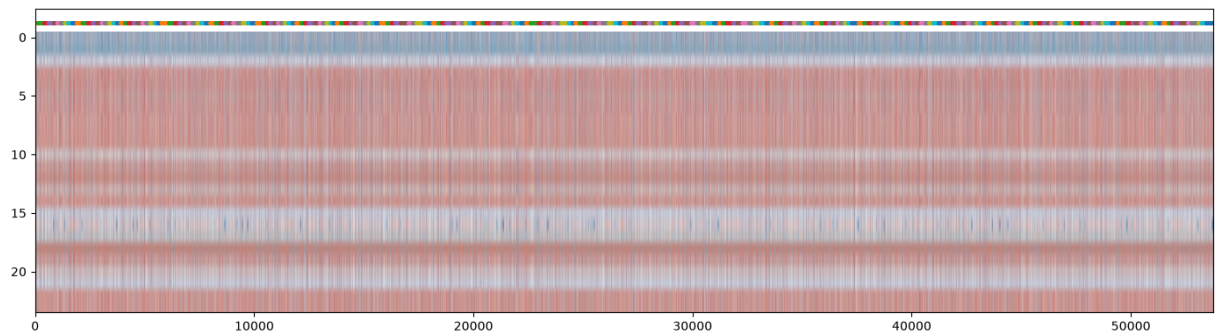


- The data contains series for multiple machines
- These are highlighted at the top with different colors

Heatmaps

Now, let's plot all parameters and sensors for *FD002*

```
In [8]: tmp = data_sv_dict['train_FD002']
        util.plot_dataframe(tmp[dt_in], labels=tmp['machine'], figsize=figsize)
```

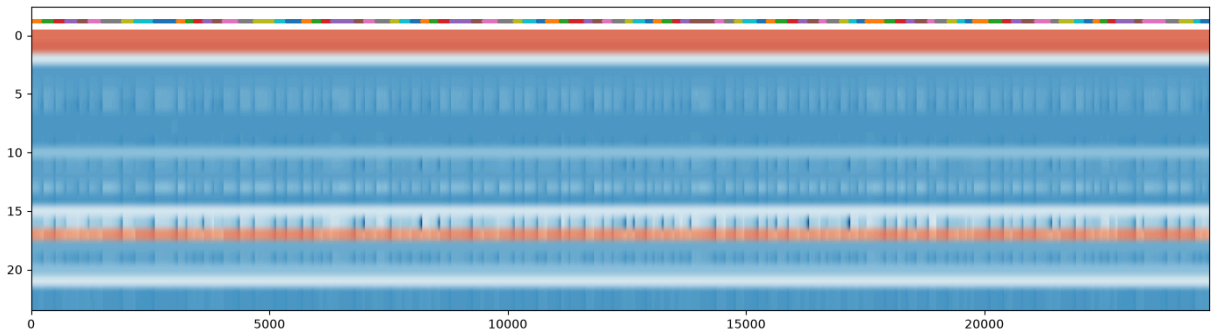


- The series is much more variable in this case
- This is due to the multiple operating conditions

Heatmaps

Now, let's plot all parameters and sensors for *FD003*

```
In [9]: tmp = data_sv_dict['train_FD003']
        util.plot_dataframe(tmp[dt_in], labels=tmp['machine'], figsize=figsize)
```

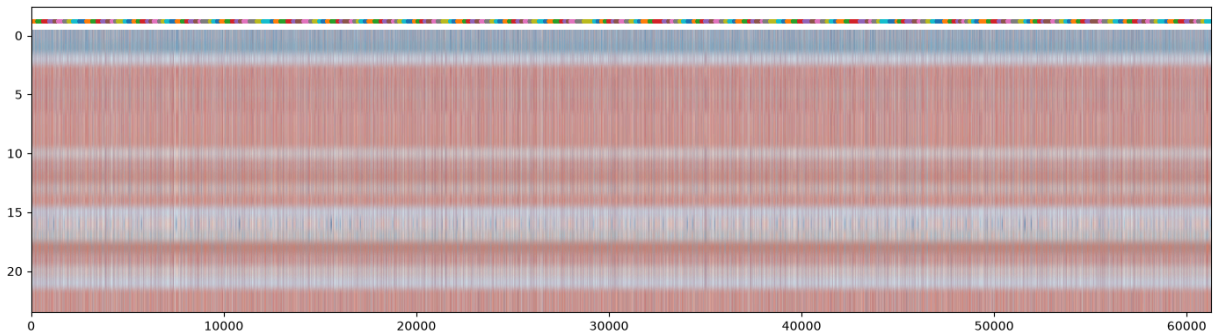


- Only one operating condition in this case (but two fault modes)
- The series is similar to FD001

Heatmaps

Finally, let's plot all parameters and sensors for *FD004*

```
In [12]: tmp = data_sv_dict['train_FD004']
util.plot_dataframe(tmp[dt_in], labels=tmp['machine'], figsize=figsize)
```

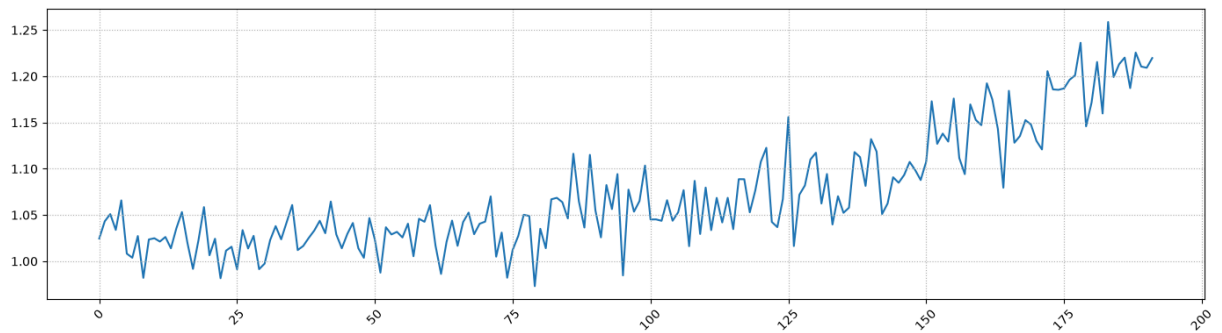


- Again six operating conditions
- ...And the series is similar to FD004

Selected Column

Let's plot one column in deeper detail *for a single machine* in *FD001*

```
In [13]: tmp = data_sv_dict['train_FD001']
tmp = tmp[tmp['machine'] == tmp['machine'].iloc[0]]
util.plot_series(tmp['s4'], figsize=figsize)
```

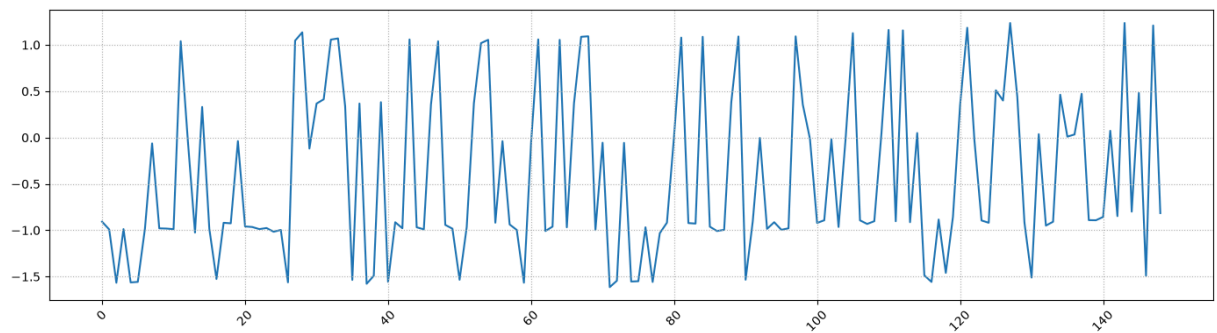


- A clear trend, possibly correlated to component wear

Selected Column

Let's see the same column for *FD002*

```
In [14]: tmp = data_sv_dict['train_FD002']
tmp = tmp[tmp['machine'] == tmp['machine'].iloc[0]]
util.plot_series(tmp['s4'], figsize=figsize)
```

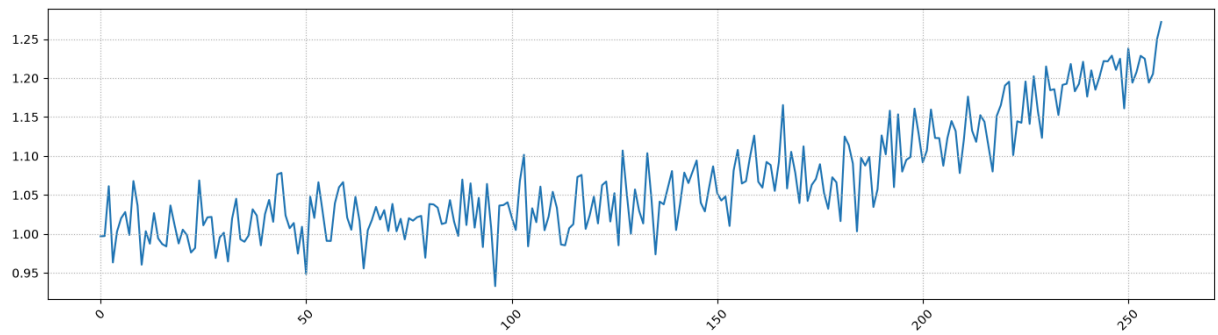


- The trend is still present, but weaker and hidden by wide oscillations

Selected Column

...And then the same column for *FD003*

```
In [15]: tmp = data_sv_dict['train_FD003']
tmp = tmp[tmp['machine'] == tmp['machine'].iloc[0]]
util.plot_series(tmp['s4'], figsize=figsize)
```

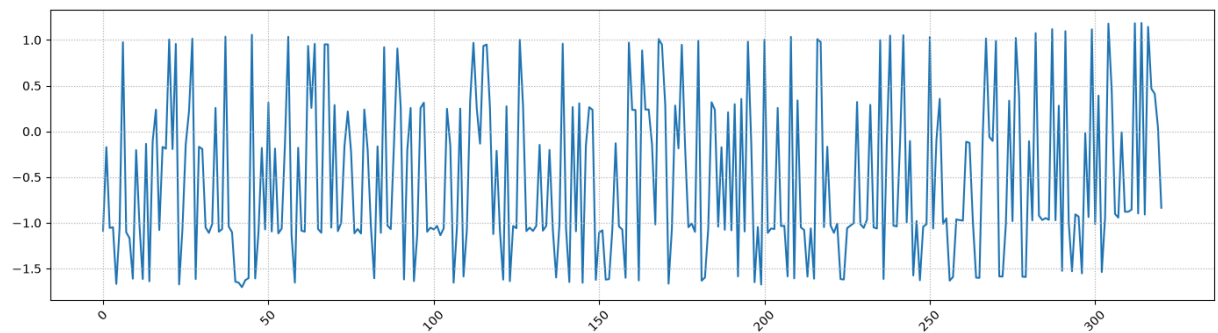


- Clear trend, with small oscillations that are more frequent than FD001

Selected Column

Let's see the same column for *FD004*

```
In [16]: tmp = data_sv_dict['train_FD004']
tmp = tmp[tmp['machine'] == tmp['machine'].iloc[0]]
util.plot_series(tmp['s4'], figsize=figsize)
```



- Very weak trend, wide and frequent oscillations